

CEUS-NET: LESION SEGMENTATION IN DYNAMIC CONTRAST-ENHANCED ULTRASOUND WITH FEATURE-REWEIGHTED ATTENTION MECHANISM

Peng Wan^{1†}, Fang Chen^{1†}, Xiaowei Zhu¹, Chunrui Liu², Yidan Zhang², Wentao Kong^{2*},
Daoqiang Zhang^{1*}

¹Nanjing University of Aeronautics and Astronautics
MIT Key Laboratory of Pattern Analysis and Machine Intelligence
(*Email: dqzhang@nuaa.edu.cn)

²Drum Tower Hospital, The Affiliated Hospital of Nanjing University Medical School
Department of Ultrasound
(*Email: breezewen@163.com)

ABSTRACT

Contrast-enhanced ultrasound (CEUS) has been a popular clinical imaging technique for the dynamic visualization of the tumor microvasculature. Due to the heterogeneous intra-tumor vessel distribution and ambiguous lesion boundary, automatic tumor segmentation in the CEUS sequence is challenging. To overcome these difficulties, we propose a novel network, CEUS-Net, which is a novel U-net network infused with our designed feature-reweighted dense blocks. Specifically, CEUS-Net incorporates the dynamic channel-wise feature re-weighting into the Dense block for adapting the importance of learned lesion-relevant features. Besides, in order to efficiently utilize dynamic characteristics of CEUS modality, our model attempts to learn spatial-temporal features encoded in diverse enhancement patterns using a multi-channel convolutional module. The CEUS-Net has been tested on tumor segmentation tasks of CEUS images from breast and thyroid lesions. It results in the dice index of 0.84, and 0.78 for CEUS segmentation of breast and thyroid respectively.

Index Terms— Contrast-enhanced ultrasound, Feature re-weighting, Tumor segmentation

1. INTRODUCTION

Contrast-enhanced ultrasound (CEUS) is a safe, inexpensive and noninvasive imaging modality that is widely applied in tumor diagnosis and prognostic evaluation[1]. By taking advantage of intravascular microbubble contrast agents (CAs), CEUS creates a significant opportunity for radiologists to visualize the dynamic enhancement process of tissue perfusion. The relative intensity changing difference between lesions

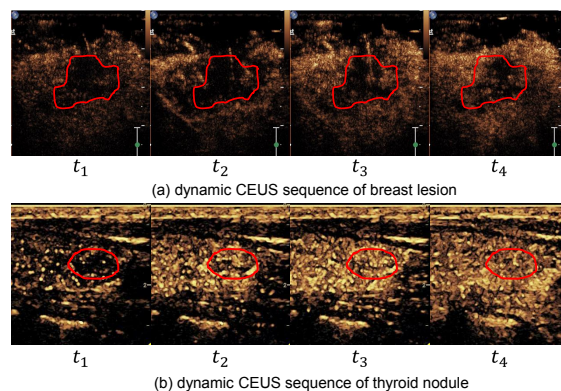


Fig. 1. Two examples of dynamic enhanced CEUS sequences. The first row presents the case of malignant breast lesion, the second is the case of thyroid nodule. For simplicity, we selected four representative images over the whole perfusion phases. Red circle outlines the tumor area.

and their adjacent normal tissues helps assess the morphological and functional characteristics of tumor microvasculature. Therefore, CEUS has been increasingly significant in lesion detection and tissue characterization. Throughout the tumor treatment, radiologists always require to locate the lesion position, quantify their morphology and size by observing CEUS sequential images back and forth[1]. However, manual segmentation in the CEUS sequence is a time-consuming and tedious task, which routinely needs to monitor the dynamic behavior of CAs across the whole perfusion. Besides, segmentation accuracy heavily depends on the knowledge and experience of clinicians, suspicious to subjective errors. Considering the increasing amount of CEUS data, automatic lesion segmentation is highly desirable to standardize the imaging evaluation.

Automatic processing in CEUS imagings is probably dif-

[†]These authors contributed equally to this work.

*Corresponding authors.

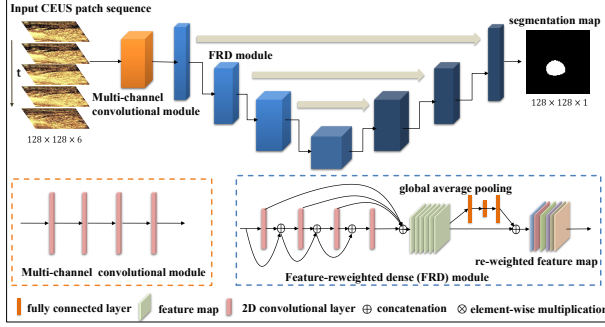


Fig. 2. The flowchart of our proposed CEUS-Net architecture.

difficult due to inter-subject variability in tumor location, shape and size. As shown in Fig.1, the interior of lesion tends to be heterogeneous as a result of different degrees of tumor angiogenesis and local necrosis. Apart from that, the lesion boundary appears to be obscure or at least incomplete for the nature of tumor invasive growth[2]. Both the two imaging characteristics pose a great challenge to the task of automatic lesion detection in CEUS modality.

In this paper, we present a feature-reweighted convolutional neural network called CEUS-Net for lesion segmentation in CEUS sequences. The flowchart is shown in Fig.2, CEUS-Net is a novel U-net network infused with our proposed feature-reweighted dense block (FRD). Specifically, the main network architecture is designed from the following perspectives, On the one hand, we incorporate the dynamic channel-wise feature-reweighted mechanism into the compact internal representations of Dense block[3], which aims at adaptively adjusting the importance of feature maps and distilling features encoding richer semantic concept. On the other hand, in order to efficiently utilize dynamic information contained in various enhancement patterns, we add a multi-channel convolutional module at the input of network with the goal to extract spatial-temporal characteristics from contiguous CEUS sequence. The performance of CEUS-Net has been tested on lesion segmentation tasks both in the breast and thyroid CEUS datasets. Experimental results validate the accuracy and effectiveness of the proposed CEUS-Net.

2. METHOD

2.1. Network architecture of CEUS-Net

As shown in Fig.2, CEUS-Net architecture is a modification of the U-net[4], a well-known deep model applied in medical image segmentation. CEUS-Net shares the same architecture as U-net with an encoder-decoder structure, separated by a bottleneck block. Given the input dynamic CEUS sequence $I_{CEUS} = \{I_0, I_1, I_2 \dots I_n\}$ with the fixed-length n , CEUS-net aims at learning a non-linear function $F_{CEUS}^{seg}(\cdot)$, that maps the CEUS series I_{CEUS} to the lesion boundary

score map S . Specifically, the feature-reweighted dense block (FRD) module is the basic building block in our proposed CEUS-Net, there are two kinds of architectures designed in the experiment, one is constructed with 5 FRD modules and the other is built on 11 FRD modules.

2.2. Multi-channel convolution module

Spatial-temporal information extraction is a critical issue in the analysis of dynamic medical imaging[5], especially considering that temporal characteristics (i.e., diverse perfusion enhancement patterns) of CEUS series are closely associated with intra-tumor angiogenesis. Thus, we introduce a multi-channel convolution module $f_{conv}^1(\cdot)$ before the U-net architecture, which allows for the input of image patch sequence $I_{CEUS} = \{I_0, I_1, I_2 \dots I_n\}$ cropped from dynamic CEUS series. This module is composed of multiple multi-channel 2D convolution layers and the numbers of channels are 6, 16, 32, 64, respectively. The output $f_{conv}^1(I_{CEUS})$ is expected to encode information of dynamic behavior of contrast agent and then passed to subsequent network for lesion segmentation.

2.3. Feature-reweighted dense (FRD) module

It is worth noting that increasing researches find that it is inappropriate to consider the feature map at each channel to be of equal importance[6]. Feature reweighting mechanism provides an effective way to differentiate the channel-wise contribution, allowing the model to focus on the most relevant part of images or features in specific tasks. By explicitly modeling the interdependencies among channels, feature reweighting mechanism can catch and exploit the relationships between separated feature channels.

Firstly, we employ a global function to embed the entire spatial information of input feature map. The input feature map $\mathcal{F}$ with size of $H \times W \times C$ is processed with the global function, $\mathcal{G} = \frac{1}{h \times w} \sum_{i=1}^H \sum_{j=1}^W \mathcal{F}(i, j)$, where $\mathcal{F}(i, j)$ is the feature value on the position (i, j) of the feature map, and $\mathcal{G}$ is a vector with a size of $1 \times C$. As studied in [6], we utilize a gating approach to capture the channel-wise dependencies from the retrieved global function. Soft-shrinkage β and sigmoid activity function σ are employed to implement the gating mechanism, as:

$$W = \beta(\mathcal{L}_U(\sigma(\mathcal{L}_D(\mathcal{G})))) \quad (1)$$

where W is learned channel specific descriptor corresponding to relative importance of channels, $\mathcal{L}_D \in R^{C \times \frac{C}{r}}$ and $\mathcal{L}_U \in R^{\frac{C}{r} \times C}$ are the channel reduction and channel up-sampling operators respectively, r is empirically set to 2.

Driven by the success of the dense-connection architecture proposed in DenseNet[3], we introduce the feature-reweighted dense (FRD) module combining feature reweighting mechanism and dense-connection design. As shown in

Fig.2, each layer in the FRD module can fully reuse information from preceding layers by connecting to the previously computed feature maps. All the generated features accumulate at the end of every Dense unit, creating a very rich set of representations. Let U_i is the output tensor of the i -th FRD module, $U_i = f_{CEUS}^i(U_{i-1})$, R_i^j is the activated feature map j -th conv+RELU layer of the i -th FRD module, U_i can be expressed as,

$$U_i = W \cdot f_i, f_i = [R_i^N; R_i^{N-1}; \dots; R_i^1; U_{i-1}] \quad (2)$$

Specially, we have chosen to include a feature reweighting mechanism only at the end of every Dense unit in order to reweight the accumulated feature-maps of all preceding layers. Thus, the variations in the information learned at each layer in the form of the features map are weighted according to their importance as determined by the loss function.

3. EXPERIMENTS AND RESULTS

3.1. Dataset and Implementation details

The training of the CEUS-Net is performed using CEUS datasets of breast and thyroid. CEUS data were collected from the local hospital and gained the approval of institutional review board. The breast and thyroid databases comprise 85 and 163 samples respectively, having the same resolution of 320*320 pixels. Considering the relatively slow blood perfusion process, we selected 6 critical points from original image sequence. Specifically, we first determined the point at which perfusion volume achieves the maximal. Then, taking the peak point as reference time, we extracted frames time proportionally, at 25%, 45%, 75%, 100%, 175%, 250% with the respective to the reference point, ensuring the selected six CEUS frames cover complete wash-in and wash-out phases. Pixel-level ground-truth of the lesion was annotated by experienced radiologists with more than 5-year clinical experience. For model training and evaluation, we randomly divided all samples into three parts, training set, validation set and test set in a ratio 4:1:1.

Our network takes in CESU patch sequence of $128 \times 128 \times 6$ out of the selected CEUS subsequence with 50% spatially overlap. At test stage, the input patch size is increased to $256 \times 256 \times 6$ and selected using 50% overlap similar to training. The network was implemented using a TensorFlow framework. Network performance is stochastically optimized using *cross-entropy loss* with RMSProp Optimizer, and an initial learning rate of 0.0001 and decay of 0.995 was used. A small batch size of 4 images is used for training. The networks are trained for 200 epochs in all experiments. The probability map of network output was converted into a binary mask by an empirical threshold value.

3.2. Network Evaluation on Lesion Segmentation in Dynamic CEUS

In this section, we present the quantitative and qualitative results of our experiments on breast and thyroid datasets. To help us validate the effectiveness of feature re-weighting mechanism for CEUS imaging segmentation, we compare the performance of our CEUS-Net with that of FC-Densenet[7] and Unet[4]. For CEUS-Net and FC-Densenet, we performed lesion segmentation on architectures with different network depths. FC-Densenet-67 and FC-Densenet-103 represent the network with 67 and 103 convolutional layers in total. CEUS-Net-v5 and CEUS-Net-v11 denote our proposed network with 5 FRD modules and 11 modules respectively.

Table 1. Quantitative evaluation results of different methods for breast and thyroid lesion segmentation

Network	breast			
	Acc(%)	F1 score(%)	Dice(%)	IoU(%)
Unet	89.20±6.41	88.67±6.16	77.96±10.73	65.30±11.27
FC-Densenet-67	89.81±5.77	89.42±5.57	79.66±9.91	67.67±10.53
FC-Densenet-103	91.07±5.71	90.78±5.21	81.45±8.62	70.44±9.17
CEUS-Net-v5	91.08±4.34	90.91±4.26	83.09±8.95	71.90±7.98
CEUS-Net-v11	92.46±4.29	92.35±4.09	84.94±6.76	74.64±6.65

Network	thyroid			
	Acc(%)	F1 score(%)	Dice(%)	IoU(%)
Unet	88.60±5.34	88.36±6.34	67.98±9.94	53.11±10.94
FC-Densenet-67	90.30±5.11	89.71±5.94	74.14±8.74	60.51±8.89
FC-Densenet-103	91.12±4.54	90.79±5.37	75.86±5.94	62.69±6.33
CEUS-Net-v5	90.99±5.27	91.19±5.81	74.25±7.94	60.06±7.10
CEUS-Net-v11	91.81±4.33	91.61±5.15	78.23±5.44	64.99±5.91

Breast lesion segmentation: Tab.1 presents the segmentation results of breast lesion on testing CEUS sequences. Its probability output is binarized using an empirically optimal threshold value 0.42 to obtain the final segmentation map. Overall, CEUS-Net can acquire appropriate segmentation results for different samples, validating its capacity to lesion-related spatial-temporal feature extraction and handle inter-subject variability in intra-tumor angiogenesis. The highest dice index 84.94% and IoU 74.64% are observed by our proposed CEUS-Net-v11. Without feature attention mechanism, FC-Densenet-103 only achieves 81.45% and 70.44%. Fig.3 shows two examples of breast lesion segmentation output. For illustration, we only selected one representative frame from original CEUS sequences.

Thyroid nodule segmentation: Apart from breast lesion segmentation, we also test our model on thyroid dataset. Similarly, we selected 0.46 empirically as binarized value. In summary, lower metric values show thyroid nodule detection is more challenging than breast lesion segmentation. This can be explained by smaller nodule size and barely distinguishable enhancement pattern between nodule and the surrounding tissues. Relatively better results are achieved by deeper network architectures CEUS-Net-v11 with 78.23% dice and FC-Densenet-103 with 75.86% dice, which vali-

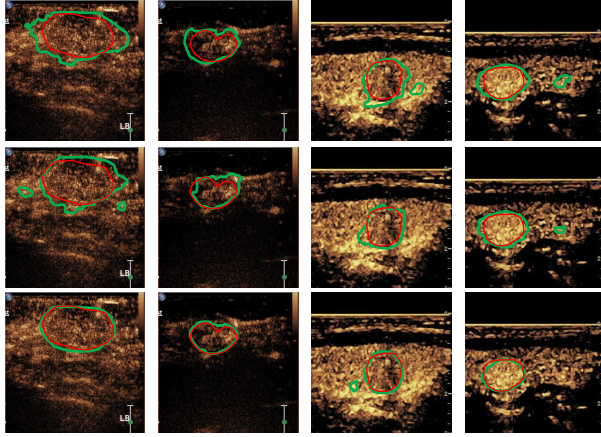


Fig. 3. Comparison of segmentation output of different methods. The results obtained by U-net, FC-Densenet-103, and CEUS-Net-v11 are shown in the first, second and third row respectively. The first two column are two examples of breast lesion and the last two column correspond to results of thyroid nodule. Red contours denote the groundtruth and green curves present the segmented regions.

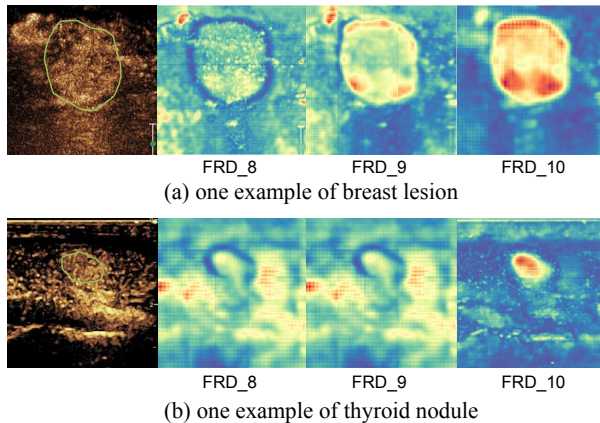


Fig. 4. Heatmap of the intermediate outputs from up-sampling path. The first column is selected CEUS frame from original sequence. The second, third and fourth column are resized heatmaps from the output of the 8th FRD, 9th FRD and 10th FRD module respectively.

dates that deeper network can possess better generalization for thyroid nodule segmentation. Fig.3 shows two examples of thyroid nodule segmentation output.

Feature maps: In addition, we extracted the concatenated output of the up-sampling path and resized it to the size of the original input. As can be observed from Fig.4, for the last few modules, the activation of the boundary area is obviously distinguished from other tissue, interpreting good segmentation results of our network. Furthermore, we can observe richer intratumor variety from the feature heatmap, which might pro-

vide us a novel perspective for tumor evaluation with CEUS imaging. It is noted that more effective temporal fusion mechanism might be explored in the future for dynamic characteristic extraction by explicitly taking the temporal dependency into account.

4. CONCLUSION

In this paper, a novel framework CEUS-Net is proposed for lesion segmentation in dynamic CEUS images. To fully leverage the dynamic characteristic of contrast agent enhancement, a multi-channel convolutional module is introduced to learn deep spatial-temporal features for subsequent lesion detection. The feature-reweighted mechanism attempts to adaptively filter effective feature maps to extract lesion relevant features. Better experimental results both on breast and thyroid datasets demonstrate the capacity of our model to discriminate lesion from normal tissues.

5. ACKNOWLEDGEMENT

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